

1- Star absorption

$$x_i^{t+1} = x_i^t + r (x_{BH} - x_i^t)$$

2-Star sucking:

$$R = \frac{f_{BH}}{\sum_{i=1}^N f_i}$$

Algorithm 2 BH Algorithm

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1: Initialize a random population of stars (  $x_i$  and  $i \in [1, N]$ )
2: Calculate the fitness of each star
3:  $x_{BH} \leftarrow$  Optimal solution
4: while not (termination criterion) do
5:   for each individual  $x_i$  and  $i \in [1, N]$  do
6:      $x_i^{t+1} \leftarrow x_i^t + r (x_{BH} - x_i^t)$ 
7:   end for
8:    $R \leftarrow \frac{f_{BH}}{\sum_{i=1}^N f_i}$ 
9:   for each individual  $x_i$  and  $i \in [1, N]$  do
10:     $D_i \leftarrow \sqrt{(x_{1BH} - x_{1i})^2 + \dots (x_{mBH} - x_{mi})^2}$ 
11:    if  $D_i < R$  then
12:       $X_i \leftarrow$  Randomly generated individual
13:    end if
14:   end for
15:   Calculate the fitness of each star
16:    $x_{BH} \leftarrow$  Optimal solution
17: end while
18: return

```

① each soln is a star.

Example: find the value for x & y that minimize $f = x^2 + y^2$ using BH algorithm.

$$\text{if } 0.5 \leq x \leq 3$$

$$1 \leq y \leq 7$$

Solution:

$$X = X_{\min} + (X_{\max} - X_{\min}) * r$$

step ①: Initialization

Stars = 5 solutions

Iter = 2

$$x_i^{t+1} = x_i^t + r (x_{BH} - x_i^t)$$

It = 1

Initialization

1- Star absorption

2-Star sucking:

$D < R$

X	y	f	x'	y'	f'	R	D	x'	y'
2.16	4.88	28.48	2.05	4.66	25.92	0.15	1.73	2.05	4.66
1.32	4.14	18.88	0.86	4.09	16.47	0.15	0.62	0.86	4.09
1.67	3.51	15.11	1.6	3.51	14.88	0.15	0.8	1.6	3.51
2.29	3.24	15.74	2.07	3.3	15.17	0.15	1.28	2.07	3.3
0.8	3.47	12.68	0.8	3.47	12.68	0.15	0	0.8	3.47

x_{BH}

$$D_i = \sqrt{(x_{BH} - x_i)^2 + (y_{BH} - y_i)^2}$$

Intilization

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$D < R$

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2.16	4.88	28.48	2.05	4.66	25.92	0.15	1.73	2.05	4.66
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1.67	3.51	15.11	1.6	3.51	14.88	0.15	0.8	1.6	3.51
2.29	3.24	15.74	2.07	3.3	15.17	0.15	1.28	2.07	3.3
0.18	3.47	12.68	0.8	3.47	12.68	0.15	0	0.8	3.47

1- Star absorption

2-Star sucking:

1- Star absorption

x^2	y^2	f^2	R	D	x^2	y^2	x^3	y^3	f^3
1.63	3.79	17.02	0.18	0.68	1.63	3.79	1.61	3.54	15.12
0.81	3.49	12.84	0.18	0.38	0.81	3.47	0.92	3.45	12.75
0.82	3.48	12.78	0.18	0.37	0.82	3.48	0.92	3.41	12.47
1.15	3.31	12.28	0.18	0	1.15	3.31	1.15	3.31	12.28
0.8	3.47	12.68	0.18	0.38	0.8	3.47	0.89	3.34	11.95



